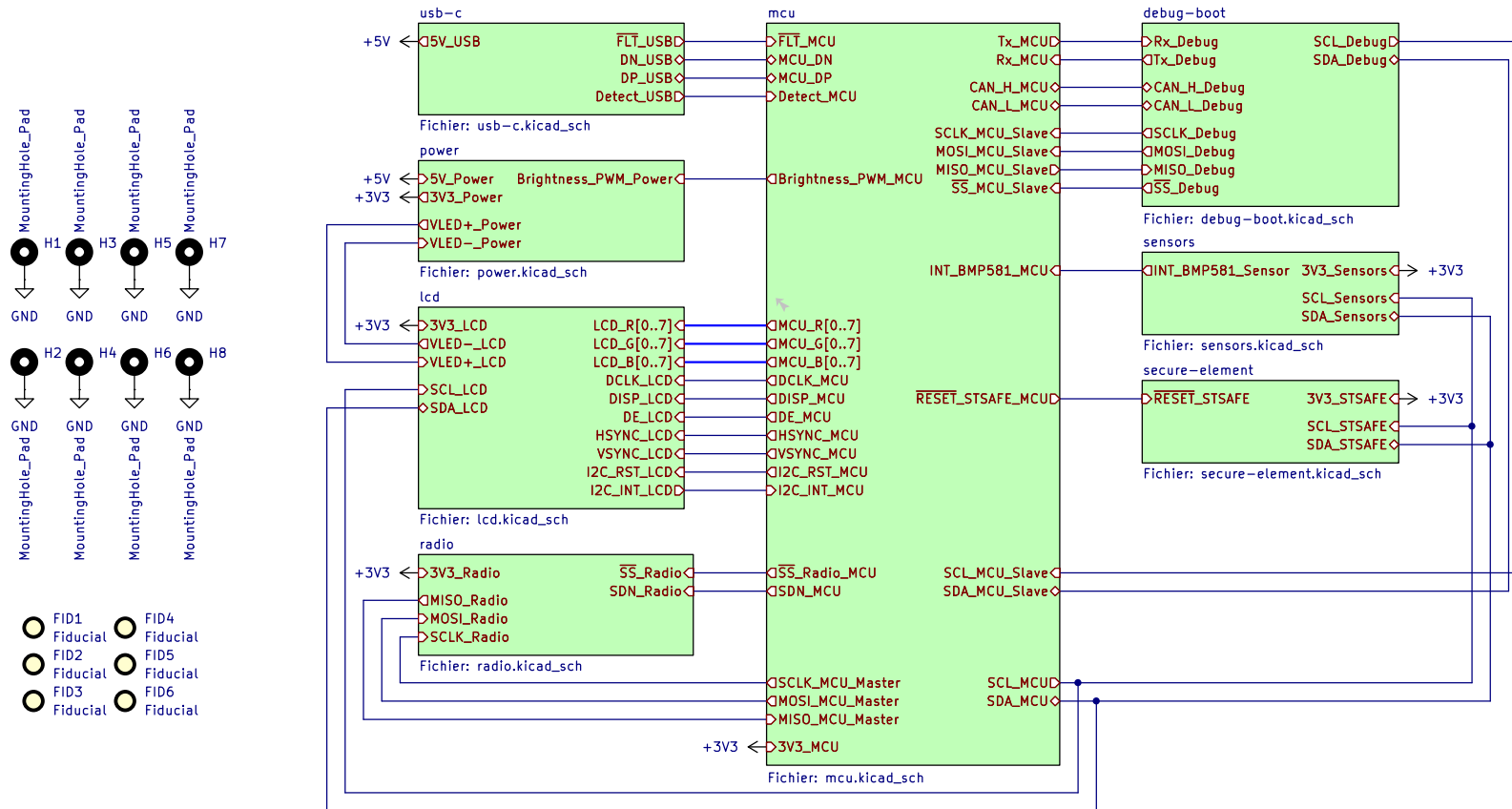


Weather station POC



block-diagram

Fichier: block-diagram.kicad_sch

Revision			
Rev.	Description	Date	Approved
A	Project creation	01/11/2025	JCR

Julien Cruvieux

Sheet: /
File: weather-station-poc.kicad_sch

Title: Weather station POC

Size: A4 Date: 2025-12-26

KiCad E.D.A. 10.0.0

Rev: A

Id: 1/10

JLCPCB stackup

Layers: 4 Thickness: 1.6 Outer Copper Weight: 1 Inner Copper Weight: 0.5

No requirement	JLC04161H-7628	JLC04161H-3313	JLC04161H-1080
JLC04161H-7628A	JLC04161H-3313A	JLC04161H-1080A	JLC04161H-7628B
JLC04161H-2116A	JLC04161H-2116B	JLC04161H-2116C	JLC04161H-2116
JLC04161H-7628E	JLC04161H-2116D	JLC04161H-7628F	JLC04161H-2116E
JLC04161H-7628D	JLC04161H-7628C		

Layer	Material Type	Thickness	
Top Layer	Copper	0.035mm	
Prepreg	7628*1	0.21040mm	
Inner Layer L2	Copper	0.0152mm	
Core	Core	1.065mm	1.1mm H/HOZ with copper
Inner Layer L3	Copper	0.0152mm	
Prepreg	7628*1	0.21040mm	
Bottom Layer	Copper	0.035mm	

7628 FR DK = 4.4

So 0.3mm trace width => 54.9 ohms

For USB differentials pairs, 0.3mm trace width and 0.3mm spacing => 96.4 ohms (90 ohms +/- 10% for USB)

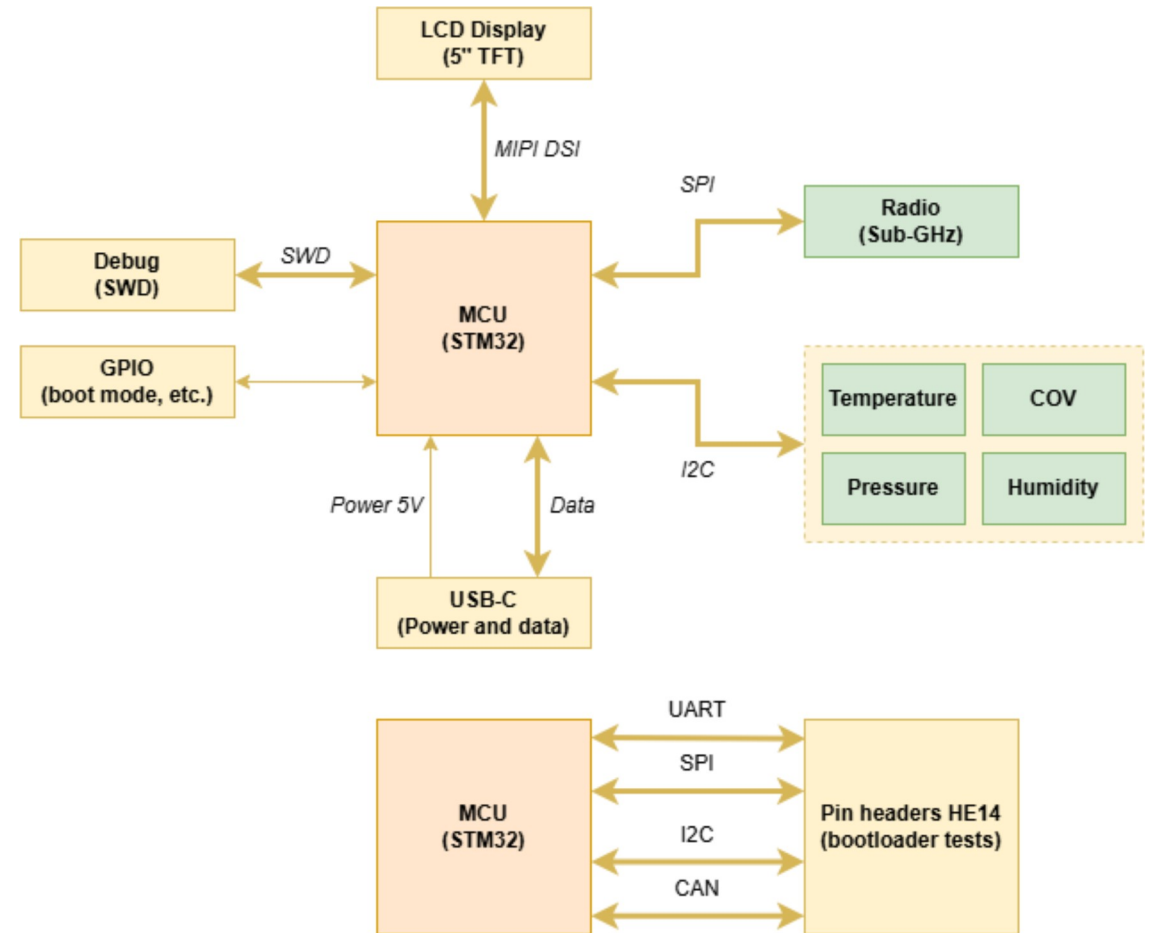
Calculation according PCB way impedance calculator :

(https://www.pcbway.com/pcb_prototype/impedance_calculator.html)

PCB build up

L1 : SIG + PWR	PP
L2 : GND	
	Core (7628 FR4)
L3 : GND	
L4 : SIG + GND	PP

L1 and L4 have a direct return paths through solid ground plane on L2 and L3. But Pwr on L1 is perhaps an issue in terms of EMC ?



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Sheet: /block-diagram/

File: block-diagram.kicad_sch

Title: Block diagram

Size: A4

Date: 2025-12-26

Rev: A

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Id: 2/10

USB-C

USB-C connector right angle

IC2 632723100011

PWR_FLAG

GND

CC1c

DP12

DN12

VBUS

VBUS_1

CC1

DP1

DN1

SBU1

VBUS_2

SSRXN2

SSRXP2

GND_2

GND_3

SSTXP2

SSTXN2

VBUS_3

CC2

DP2

DN2

SBU2

VBUS_4

SSRXN1

SSRXP1

GND_4

GND_5

GND_6

MH1

MH2

MH3

MH4

MP1

MP2

B5

CC2c

B6

DP12

B7

DN12

B8

B9

VBUS

B10

B11

B12

G1

G2

MH1

MH2

MH3

MH4

MP1

MP2

USB-C power sink, 5V, 15W

Q2 STL11N3LLH6

PWR_FLAG

CC1c

DP12

DN12

VBUS

VBUS_1

CC1

DP1

DN1

SBU1

VBUS_2

SSRXN2

SSRXP2

GND_2

GND_3

SSTXP2

SSTXN2

VBUS_3

CC2

DP2

DN2

SBU2

VBUS_4

SSRXN1

SSRXP1

GND_4

GND_5

GND_6

MH1

MH2

MH3

MH4

MP1

MP2

B5

CC2c

B6

DP12

B7

DN12

B8

B9

VBUS

B10

B11

B12

G1

G2

MH1

MH2

MH3

MH4

MP1

MP2

ESD protection on USB-C data lines

IC1 USBLC6-2SC6

DN12

GND

DP12

I/O1_1

GND

I/O2_1

I/O1_2

VBUS

I/O2_2

DN_USB

DP_USB

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Sheet: /usb-c/

File: usb-c.kicad_sch

Title: USB C

Size: A4

Date: 2025-12-26

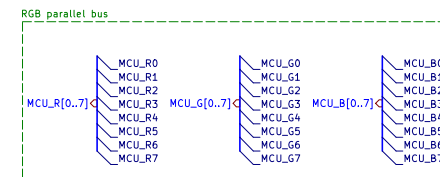
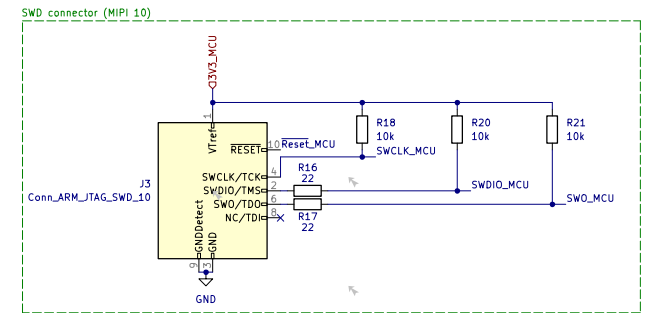
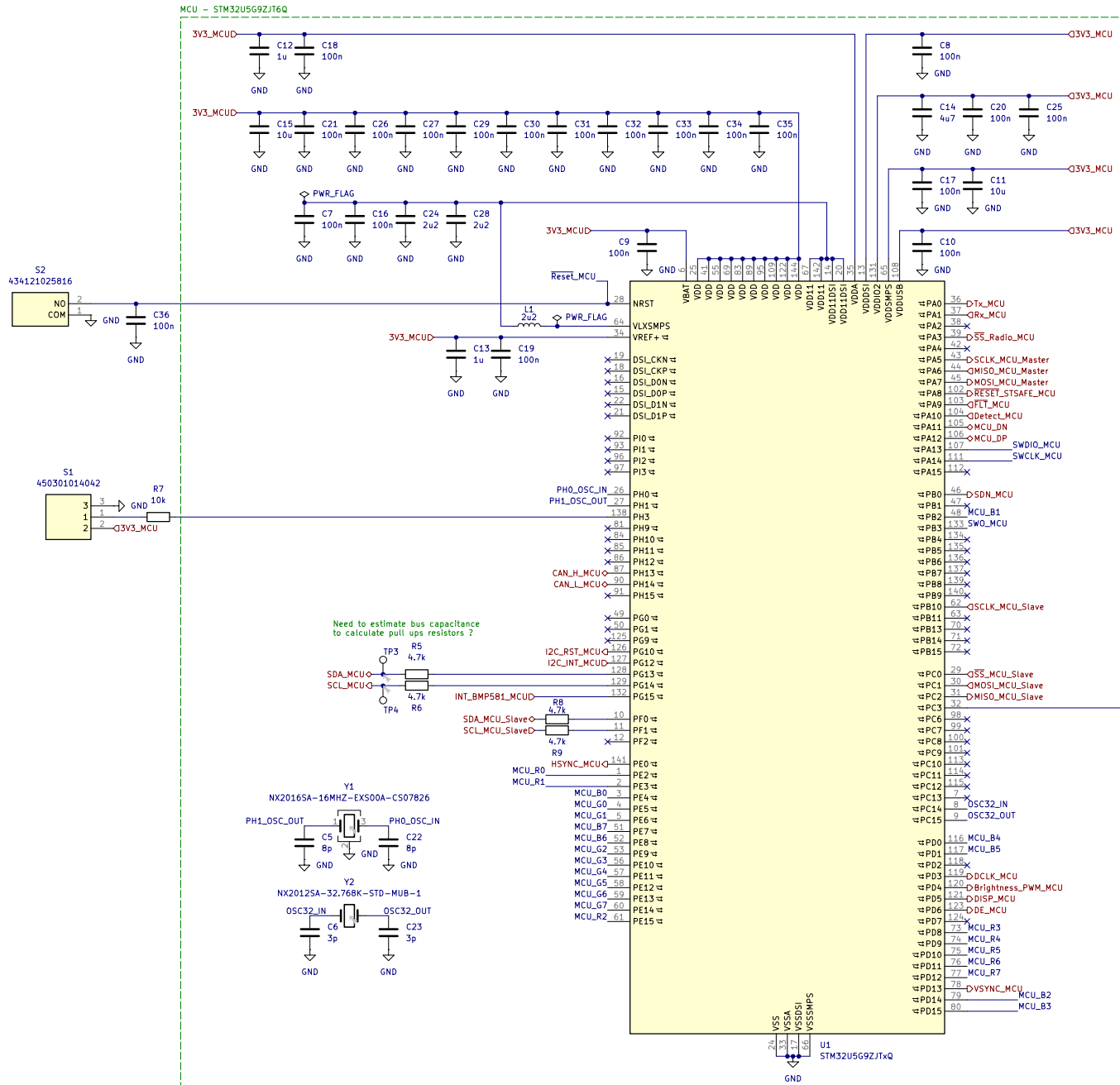
Rev: A

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Id: 3/10

Id: 3/10

MCU

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Sheet: /mcu/
File: mcu.kicad_sch

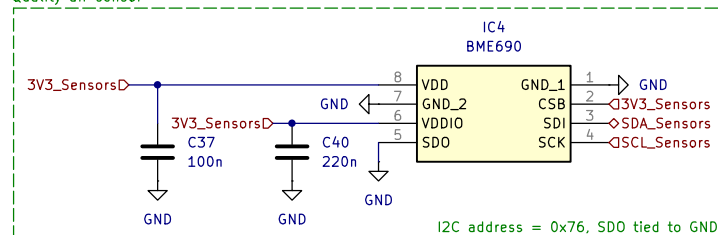
Title: MCU

Size: A3	Date: 2025-12-26
KiCad E.D.A. 10.0.0	

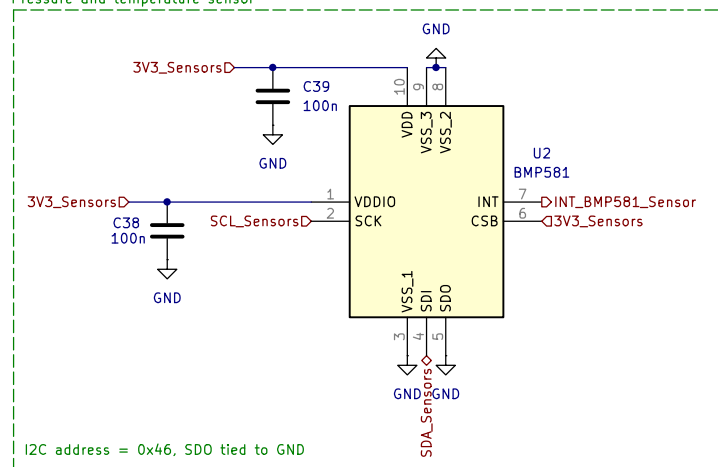
Rev: A
Id: 4/10

Sensors

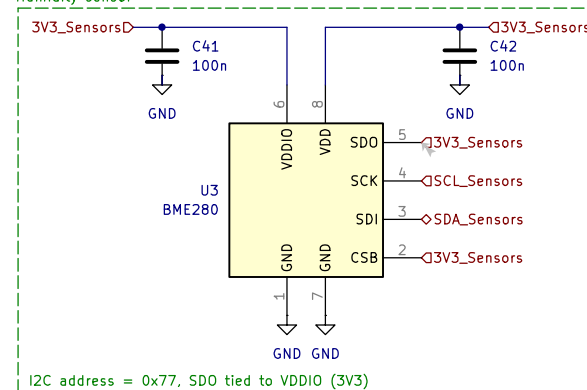
Quality air sensor



Pressure and temperature sensor



Humidity sensor



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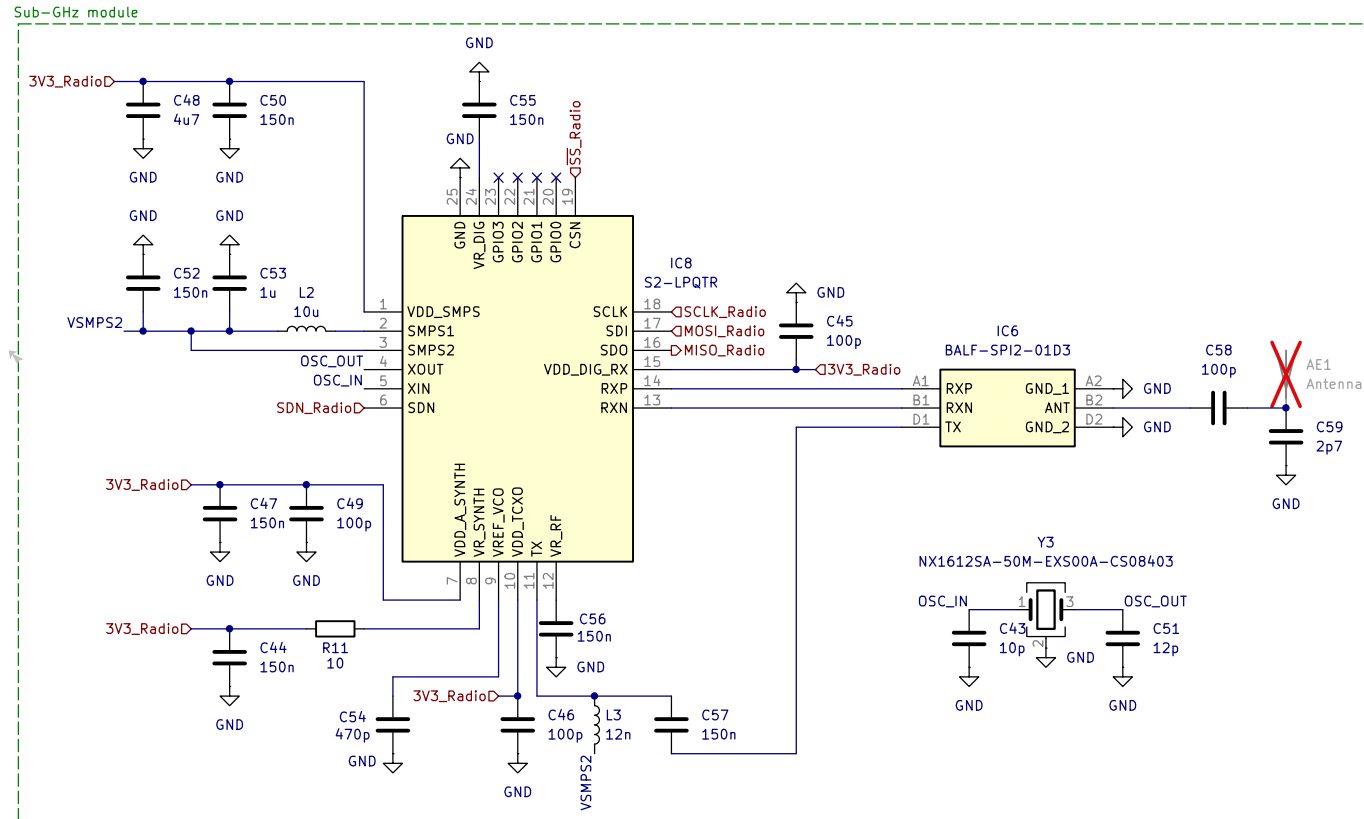
Sheet: /sensors/
File: sensors.kicad_sch

Title: Sensors

Size: A4 Date: 2025-12-26
KiCad E.D.A. 10.0.0

Rev: A
Id: 5/10

Sub-GHz radio



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Sheet: /radio/
File: radio.kicad_sch

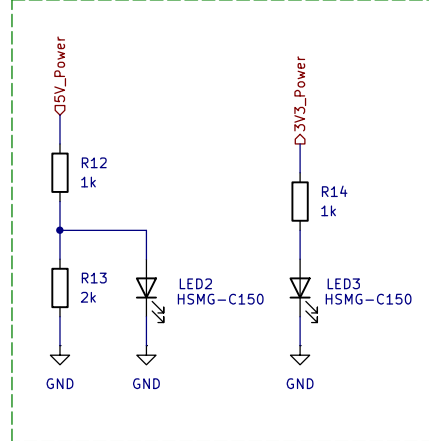
Title: Radio Sub-GHz

Size: A4 Date: 2025-12-26
KiCad E.D.A. 10.0.0

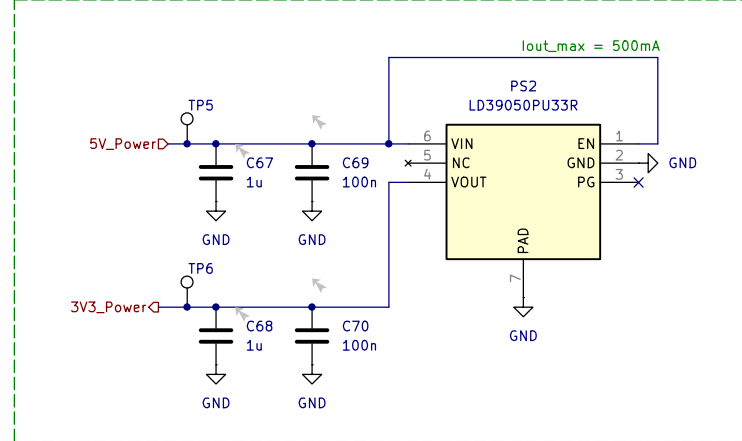
Rev: A
Id: 6/10

Power supplies

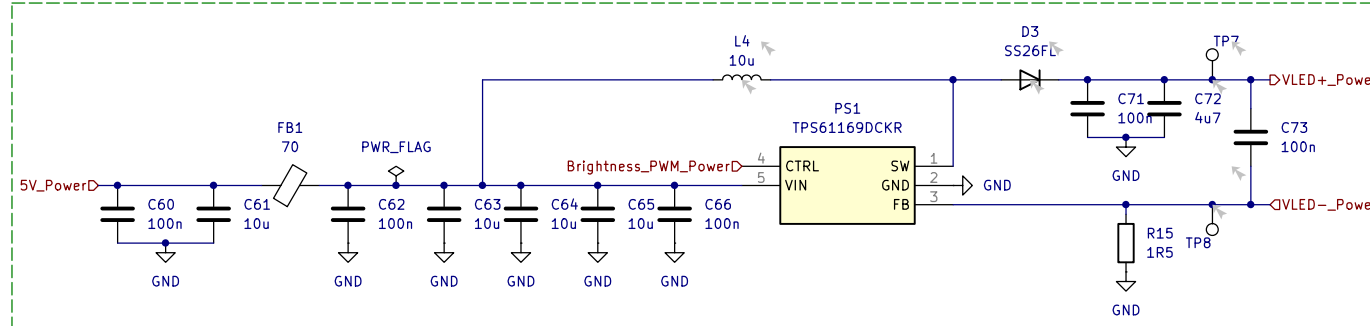
Power LED status



LDO 5V -> 3.3V



LED driver



The TPS61169 WLED (White LED) driver is a boost converter regulated in current through FB pin. The driver uses a 204mV voltage reference and the resistor on FB pin set the current threshold $\rightarrow 204\text{mV} / 1.5\text{ ohm} = 136\text{ mA}$. To have full brightness on 5" Riverdi screen, 140 mA are needed. To control the brightness, a PWM signal is injected on CTRL pin. 100% duty cycle is full brightness, 50% half brightness and so on ...

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Sheet: /power/
File: power.kicad_sch

Title: Power supplies

Size: A4 Date: 2025-12-26

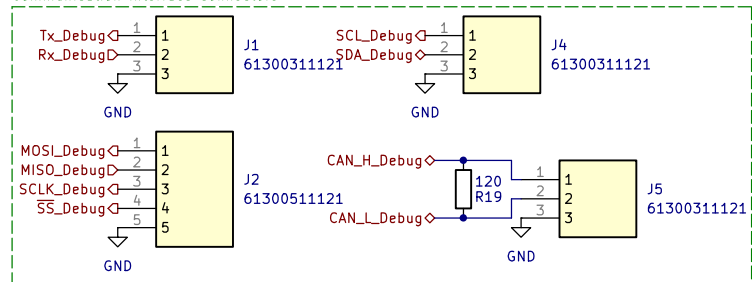
KiCad E.D.A. 10.0.0

Rev: A

Id: 7/10

Debug

Communication interface connectors



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Sheet: /debug-boot/
File: debug-boot.kicad_sch

Title: Debug

Size: A4 Date: 2025-12-26

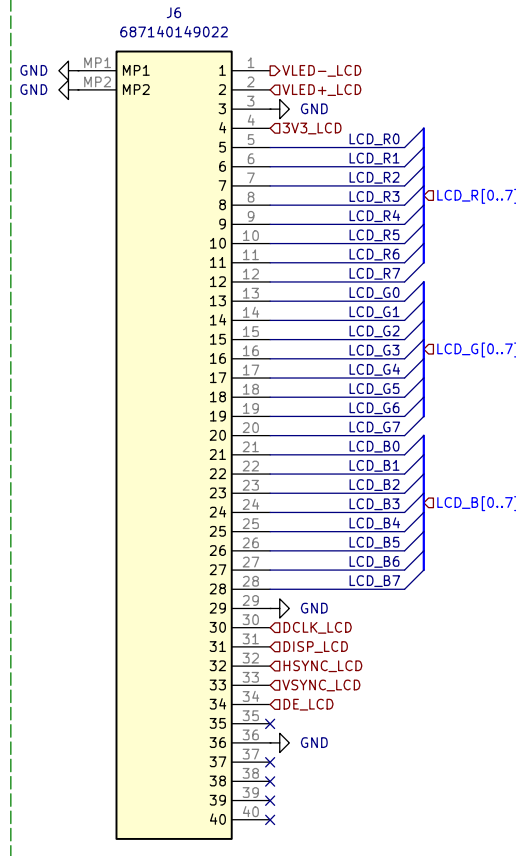
KiCad E.D.A. 10.0.0

Rev: A

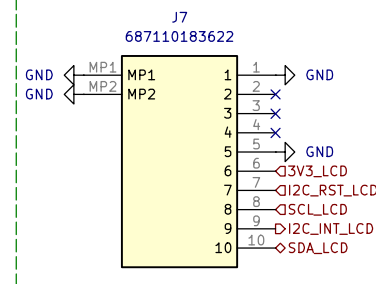
Id: 8/10

TFT LCD screen

TFT LCD 40 pins driver connector



TFT LCD 10 pins touch connector



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Sheet: /lcd/
File: lcd.kicad_sch

Title: LCD

Size: A4

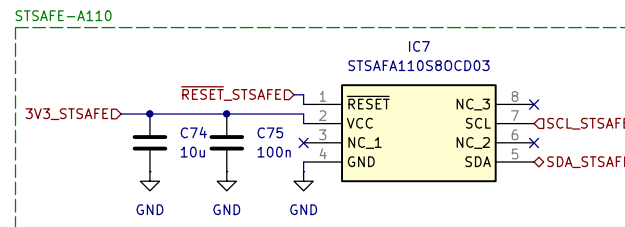
Date: 2025-12-26

Rev: A

KiCad E.D.A. 10.0.0

Id: 9/10

Secure element (STSAFE)



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Sheet: /secure-element/
File: secure-element.kicad_sch

Title: Secure elements

Size: A4 Date: 2025-12-26

KiCad E.D.A. 10.0.0

Rev: A

Id: 10/10